Professional Self-Assessment

Introduction

I am a student completing my bachelor's degree in computer science at Southern New Hampshire University. I have developed skills in software engineering and design, algorithms and data structures, and databases. This ePortfolio is a demonstration of my ability to work in collaborative environments and communicate efficiently. My showcased work demonstrates competency in problem solving, ability to overcome challenges through my enhancement work, apply innovative techniques and skills in software design and development, and mitigate safety risks through secure coding. This aligns with my goals of becoming a software developer or data analyst by applying coding and data security best practices.

Skills obtained throughout the Computer Science program

I have obtained skills for collaborating in team environments by applying agile methodologies within the software development lifecycle. I used version control by writing and pushing module assignments to GitHub. I demonstrated teamwork by participating in peer support and open discussion within the student learning portal at the university. Projects where these skills were applied as the Travlr fullstack web application and other project listed throughout this document.

Communicating with stakeholders by writing use cases, wire frames, software design templates, white pages, informal code reviews, and applying client requirements in projects. Projects this applied to were artificial intelligence neural networks and GDPR principles outlined in the white pages, the Bookhaven bookstore web application, and the Travlr website where I would incorporate professor feedback to the final product.

Software engineering and design developed core skills in developing software in languages such as C++, Python, Java, JavaScript, and SQL. Applying Object-Oriented principles to software applications. I developed skills in QA testing and automation within Junit demonstrating my ability to work with different tools. Building and analyzing AI models through a lens of an AI engineer and scientist. Incorporate UI/UX principles within the android inventory tracking application and the Bookhaven Bookstore website; among other projects that allowed me to create a fully functional website, android, and banking applications.

Data structures and algorithms through database work where I utilized sorting methods to filter data and additionally displayed the information on a visual dashboard for the Grazioso Salavare Dashboard project. Demonstrating competency with implementation of O notation, time complexities, and optimization of queries.

Database management skills through analytical problem solving by researching data within databases through written queries with the ABCU_Advising project. Improving scalability and performance by incorporating MongoDB in the Travlr website. Building tables in an SQLite database and CRUD operations within the inventory tracking application.

Security through creation of a certificate with the certificate authority, running and writing reports on security vulnerabilities within a software program, applying input validation and sanitation within code. Applying secure coding techniques to the written queries within the database to prevent SQL injection. Practicing a security mindset within each stage of the development lifecycle. With projects such as Artemis financial, contact service application, inventory tracking application, and other projects.

These skills demonstrate my ability to work in fast paced environments and learn quickly through accelerated courses offered at Southern New Hampshire University, while reinforcing the value I place on adaptability, security, communication, curiosity to finding solutions to complex problems, and a commitment to lifelong learning.

My Artifact

My artifact is an android inventory tracking application that applies all required enhancements for the capstone project. The application prior to the enhancements provided a database for storing login credentials and inventory items. It has CRUD operations to create new items, read inventory items from a grid view, update an item by quantity, and delete items from the database. It has permissions for the user to receive SMS messages when an inventory item is low in stock. Some problems this application had were a lack of security measures for SMS permissions, incomplete input validation when adding new items, and an inability to search or filter data leading to a decline in UI/UX design. To address these issues, I implemented a sorting mechanism to filter items by item name alphabetically, from quantity high to low, and quantity low to high. In addition to adding a search feature to look up items by name, input validation for adding new items to the database, and removal of the SMS feature due to changing requirements.

This project demonstrates my computer science skills by being able to apply technical proficiency with MVC architecture in android development and apply best practices, apply data management and security through an SQLite database, and adhere to UI/UX design principles with the RecyclerView, materials, and layouts. Each enhancement benefited the artifact as a whole by applying input validation, sorting features, and a search mechanism that all provide data manipulation and control of data and how it is accessed and displayed to the user. Enhancement one prevented duplicated items from being displayed on the user interface and being added to the database. Enhancement two providing sorting features to organize data and display a structural view of the inventory items to the user. Enhancement three provided searching capabilities to filter data from the database making it easier for the user to find items.

This portfolio demonstrates my ability to develop a software application that adhere to industry standards. I provide documentation and reflection throughout my work within the narratives. Lastly, through the inventory tracking application, I demonstrate these skills through practical implementation.